

Breast Cancer Classification Using Machine Learning

Baseline supervised learning pipeline using the Breast Cancer Wisconsin Diagnostic Dataset

Abstract

This project implements a reproducible supervised machine learning workflow for classifying breast cancer samples as benign or malignant using structured diagnostic feature data. The aim is to demonstrate a complete baseline pipeline covering dataset loading, exploratory analysis, preprocessing, model training, evaluation, and reporting. A Logistic Regression model is trained using a stratified train-test split and evaluated using standard classification metrics and a confusion matrix.

Dataset

The project uses the Breast Cancer Wisconsin Diagnostic Dataset available through scikit-learn. The dataset contains 569 samples and 30 numerical diagnostic features computed from digitized images of fine needle aspirate samples of breast masses. Each sample is labelled as malignant or benign.

Property	Value
Dataset	Breast Cancer Wisconsin Diagnostic Dataset
Source	scikit-learn built-in dataset
Samples	569
Features	30 numerical diagnostic features
Classes	malignant, benign
Class distribution	malignant: 212, benign: 357

Methodology

- Load the dataset using scikit-learn and convert it into a pandas DataFrame.
- Inspect sample count, feature count, class labels, class distribution, and feature summary statistics.
- Split the dataset into training and test sets using a stratified 80/20 split with random_state=42.
- Scale numerical features using StandardScaler to support stable Logistic Regression training.
- Train a Logistic Regression baseline model using a scikit-learn Pipeline.
- Evaluate the model on the held-out test set and save metrics and figures to the results directory.

Implementation Structure

The repository is organized into reusable Python scripts, an exploratory notebook, generated result files, and a project report. The source code separates dataset loading, model training, metric inspection, and single-sample prediction into separate files for clarity.

File or folder	Purpose
notebooks/01_exploration_and_baseline.ipynb	Exploratory analysis and baseline modelling walkthrough
src/load_data.py	Loads the dataset and returns features, labels, and class names
src/train_model.py	Trains the Logistic Regression baseline and saves outputs

src/evaluate_model.py	Reads and prints saved evaluation metrics
src/predict.py	Runs a single-sample prediction example
results/baseline_metrics.json	Saved evaluation metrics
results/class_distribution.png	Class distribution figure
results/confusion_matrix.png	Confusion matrix figure

Model and Evaluation

The baseline model is Logistic Regression. It is selected as the first model because it is efficient, widely used for binary classification, and easier to interpret than more complex non-linear models. The model is implemented as a scikit-learn Pipeline: StandardScaler followed by LogisticRegression.

Evaluation Metrics

The model is evaluated using accuracy, precision, recall, F1-score, ROC-AUC, and a confusion matrix. In medical classification, accuracy alone is insufficient. The confusion matrix and class-specific errors are important because a malignant case predicted as benign would be clinically serious.

Metric	Value
Accuracy	0.9825
Precision	0.9861
Recall	0.9861
F1-score	0.9861
ROC-AUC	0.9954

Note: the saved metrics follow the implementation in `src/train_model.py`. In scikit-learn binary classification, precision, recall, and F1-score use label 1 as the positive class by default, which corresponds to benign in this dataset. For clinical interpretation, the confusion matrix should also be inspected to understand malignant-class errors.

Confusion Matrix

	Predicted malignant	Predicted benign
Actual malignant	41	1
Actual benign	1	71

On the held-out test split, the baseline produced one malignant sample predicted as benign and one benign sample predicted as malignant. This indicates strong initial performance on this curated dataset, while still highlighting the importance of false negative analysis in clinical AI workflows.

Clinical Relevance and Limitations

This project demonstrates a technical baseline workflow for cancer-related machine learning using structured diagnostic features. It is useful as an introductory portfolio project because it covers the essential machine learning lifecycle: data handling, exploration, modelling, evaluation, result saving, and reporting.

However, this is not a clinically deployable diagnostic system. The dataset is relatively small, curated, and does not include raw radiology images, longitudinal patient records, genomic data, or multi-centre external validation. Real-world clinical deployment would require larger and more diverse datasets, careful bias assessment, regulatory review, prospective validation, and integration with clinician oversight.

Future Work

- Add cross-validation to estimate robustness across multiple data splits.
- Compare Logistic Regression with Random Forest, Support Vector Machine, and gradient boosting baselines.
- Add ROC curve visualization and threshold analysis.
- Add SHAP-based feature interpretation for explainability.

- Perform dedicated false negative analysis for malignant cases.
- Extend the workflow to breast ultrasound, mammography, MRI, or radiomics feature datasets.

Conclusion

This project establishes a complete first-stage machine learning portfolio project for breast cancer classification. It provides a clean and reproducible baseline that can be extended toward explainable clinical AI, medical imaging, and radiomics-focused cancer research.